

Duc Vu

✉ duchongvu01@gmail.com

🐙 [danielvu-31.github.io](https://github.com/danielvu-31)

🎓 [Google Scholar](#)

EDUCATION

Rutgers University - New Brunswick, NJ

Ph.D. in Computer Science

Advisor: Prof. Dimitris N. Metaxas

2026 - Expected: 2031

Miami University - Oxford, OH

Bachelor of Science, Data Science and Statistics

Bachelor of Arts, Economics

Honors: Cum Laude

2019 - 2024

GPA: 3.83/4.0

PUBLICATIONS

Conference Publications

* - equal contribution

6. **Cross-Space Distillation: Teaching One-Step Students with Modern Diffusion Teachers.**

Duc Vu*, Anh Nguyen*, Ngan Nguyen*, Trung Tuan Dao, Viet Nguyen, Quan Dao, Kien Nguyen, Chi Tran, Phong Nguyen, Khoi Nguyen, Cuong Pham, Dimitris N. Metaxas, Vishal M. Patel, and Anh Tran.

European Conference on Computer Vision (ECCV), 2026.

5. **InverFill: One-Step Inversion for Enhanced Few-Step Diffusion Inpainting.**

Duc Vu*, Kien Nguyen*, Trong-Tung Nguyen*, Ngan Nguyen*, Phong Nguyen, Khoi Nguyen, Cuong Pham and Anh Tran.

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

4. **Anti-I2V: Safeguarding your photos from malicious image-to-video generation.**

Duc Vu, Anh Nguyen, Chi Tran and Anh Tran.

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

3. **Improved Training Technique for Shortcut Models.**

Anh Nguyen*, Viet Nguyen*, Duc Vu, Trung Tuan Dao, Chi Tran, Toan Tran and Anh Tran.

Advances in Neural Information Processing Systems (NeurIPS), 2025.

2. **EFHQ: Multi-purpose ExtremePose-Face-HQ dataset.**

Duc Vu*, Trung Tuan Dao*, Cuong Pham, Anh Tran.

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

1. **SwiftBrush v2: Make Your One-step Diffusion Model Better Than Its Teacher.**

Trung Tuan Dao, Thuan Hoang Nguyen*, Thanh Le*, Duc Vu*, Khoi Nguyen, Cuong Pham and Anh Tran.

European Conference on Computer Vision (ECCV), 2024.

WORK EXPERIENCE

Qualcomm AI Research

AI Research Resident

Advisors: Dr. Anh Tran

March, 2025 - Current

Hanoi, Vietnam

- Developed Cross-Space Distillation, enabling step distillation from large SD 3.5 and FLUX.2 Klein models to compact one-step SD 1.5 students and boosting HPSv3 from 5.4 to 9.4.
- Developed InverFill, a one-step inversion network that enables high-fidelity, few-step inpainting with text-to-image models, matching specialized inpainting models with only 0.07 seconds of overhead.
- Introduced Anti-I2V, a dual-space defense across UNet-, DiT-, and MM-DiT-based video models, reducing identity preservation by 43% and video quality by 52%.
- Improved Shortcut Models through architectural and training refinements, achieving FID 5.27 in one step and FID 2.05 in four steps on ImageNet 256×256 .

VinAI Research

AI Research Resident

Advisors: Dr. Anh Tran

March, 2022 - March, 2025

Hanoi, Vietnam

- Engineered a dataset of 30,000 augmented infrared facial images using a 3D Morphable Model, boosting driver facial landmark detection accuracy by 15% for driver monitoring system (DMS).

- Created EFHQ, a dataset of 450,000 high-quality extreme-pose face images that improved cross-dataset GAN synthesis FID from 8.26 to 7.04, reduced reenactment pose and identity errors by up to 49% and 22%, and exposed 5–37% performance drops in state-of-the-art face recognition models.
- Developed SwiftBrush v2 using improved initialization, efficient LoRA training, a novel clamped CLIP loss and model merging, achieving a then state-of-the-art one-step FID of 8.14.

PIXTA Co. Ltd
Data Scientist Intern

December, 2020 - July, 2021
Hanoi, Vietnam

- Synthesized stock-quality images with StyleGAN2 and experimented with super-resolution methods, generating more than 2000 stock-quality images.
- Designed a face detection model on image patches specialized for high-resolution images, improving reviewing speed by around 30% and publishing time by 50% (from 7 days to 4 days).
- Deployed and evaluated a multi-task model that helped enhance customer experience, cleaned around 4 million unlabeled images (5% of total images), and re-categorized age and emotion groups of 13.5 million stock images.

SELECTED AWARDS AND HONORS

- Outstanding Resident Research Award - Qualcomm AI Research *2026*
- Presidential Fellowship - Rutgers University, New Brunswick *2026-2029*
- Scholar Leaders - Miami University, Oxford *2022-2023*
- President’s List - Miami University, Oxford *Fall 2019-20, Spring 2022-23*
- Dean’s List - Miami University, Oxford *Spring 2019-20, Spring 2021-22, Fall 2022-23, Fall 2023-24*

PROFESSIONAL RESPONSIBILITIES

- *Reviewer:* NeurIPS (2026), ECCV (2026), CVPR (2026), CVPRW (2025), ACCV (2025).

REFERENCES

Prof. Dimitris N. Metaxas
Board of Governors, Distinguished Professor, Department of Computer Science, Rutgers University
Director, Center for Computational Biomedicine, Imaging and Modeling (CBIM)
dnm@cs.rutgers.edu

Dr. Anh Tran
Principal Research Engineer, Qualcomm AI Research, Vietnam
anhtra@qti.qualcomm.com

Assoc. Prof. Cuong Pham
Dean, Faculty of Computer Science & Engineering, Posts and Telecommunications Institute of Technology (PTIT), Vietnam
Director, PTIT.AI Research Lab, Posts and Telecommunications Institute of Technology (PTIT), Vietnam
Visiting Research Scientist, Qualcomm AI Research, Vietnam
cuongpv@ptit.edu.vn

Dr. Phong Nguyen
Senior Research Engineer, Qualcomm AI Research, Vietnam
phongnh@qti.qualcomm.com